

COMM1180 Value Creation

Final Exam Formula Sheet

$$\text{Breakeven point} \implies \text{Total Sales Quantity} = \frac{\text{Total fixed costs}}{\text{Contribution margin per unit}}$$

$$\text{Target profit} \implies \text{Total Sales Quantity} = \frac{\text{Total fixed costs} + \text{Target pretax profit}}{\text{Contribution margin per unit}}$$

$$\text{Breakeven point} \implies \text{Total Sales Quantity} = \frac{\text{Total fixed costs}}{\text{Weighted average contribution margin per unit}}$$

$$\text{Target profit} \implies \text{Total Sales Quantity} = \frac{\text{Total fixed costs} + \text{Target pretax profit}}{\text{Weighted average contribution margin per unit}}$$

$$\text{Margin of safety} = \text{Actual or expected units of activity} - \text{Units at BEP}$$

$$FV = PV(1 + r)^n$$

$$PV = \frac{FV}{(1 + r)^n}$$

$$r = \left(\frac{FV}{PV} \right)^{1/n} - 1$$

$$n = \frac{\ln \frac{FV}{PV}}{\ln(1 + r)}$$

$$EAR = \left(1 + \frac{APR}{m} \right)^m - 1$$

$$APR = m \left[(1 + EAR)^{1/m} - 1 \right]$$

$$PV = C \left(\frac{1 - (1 + r)^{-n}}{r} \right)$$

$$FV = C \left(\frac{(1 + r)^n - 1}{r} \right)$$

$$C = \frac{PV}{\left[\frac{1 - (1 + r)^{-n}}{r} \right]}$$

$$PV = C \left(\frac{1 - (1 + r)^{-n}}{r} \right) (1 + r)$$

$$FV = C \left(\frac{(1 + r)^n - 1}{r} \right) (1 + r)$$

$$PV = C \left(\frac{1 - (1 + r)^{-n}}{r} \right) (1 + r)^{-k}$$

$$PV = \frac{C}{r}$$

$$PV = \frac{C}{r} (1 + r)$$

$$PV = \frac{C}{r} (1 + r)^{-k}$$

$$NPV = CF_0 + \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots$$

$$r_e = \frac{D_1}{P_0} + \frac{P_1 - P_0}{P_0}$$

$$0 = CF_0 + \frac{CF_1}{(1+\tilde{r})^1} + \frac{CF_2}{(1+\tilde{r})^2} + \frac{CF_3}{(1+\tilde{r})^3} + \dots$$

$$r_e = \frac{D_1}{P_0} + g$$

$$PI_i = \frac{NPV_i}{Cost_i} \quad \widetilde{PI}_i = \frac{PV_i}{Cost_i} \quad g_t = \text{Retention Rate}_{\%,t-1} \times \text{Return on new investments}_{\%,t}$$

$$g_t = RR_{t-1} \times ROI_t$$

$$C_{EAA} = \frac{NPV}{AF_n} = \frac{NPV}{\frac{1 - (1+r)^{-n}}{r}}$$

$$R_{t+1} = \frac{P_{t+1} - P_t + Div_{t+1}}{P_t}$$

$$r_{ARR} = \frac{\text{Ave Acc Profit}}{\text{Upfront Cost}}$$

$$\beta_i = \rho_{i,M} \frac{\sigma_i}{\sigma_M}$$

$$P_0 = C \left(\frac{1 - (1+r)^{-n}}{r} \right) + \frac{F}{(1+r)^n}$$

$$R_E = \frac{Div_1}{P_0} + g = \frac{Div_0(1+g)}{P_0} + g$$

$$PV_0 = \frac{C}{r}$$

$$C_2 = (1+g)C_1 \dots C_n = (1+g)^{n-1}C_1$$

$$E[R_i] = R_f + \beta_i(E[R_M] - R_f)$$

$$PV_0 = \frac{C_1}{r - g}$$

$$MV_{ND} = MV_D - \text{Cash}$$

$$PV_0 = \frac{C_0}{r - g} (1 + g)$$

$$V = MV_E + MV_P + MV_D$$

$$R_{WACC} = \frac{MV_E}{V} R_E + \frac{MV_P}{V} R_P + \frac{MV_D}{V} R_D (1 - T_C) = w_E R_E + w_P R_P + w_D R_D (1 - T_C)$$